

5	(a)	(i) 1	number of oscillations per unit time (not per second)	B1	[1]
		2	$n\lambda$	A1	[1]
	(ii)	$v = \text{distance} / \text{time} = n\lambda/t$		M1	
		$n/t = f$ hence $v = f\lambda$		A1	
		or f oscillations per unit time so $f\lambda$ is distance per unit time		M1	
		distance per unit time is v so $v = f\lambda$		A1	[2]
	(b)	(i)	1.0 period is $3 \times 2 = 6.0$ ms	C1	
			frequency = $1/(6 \times 10^{-3}) = 170$ Hz	A1	[2]
	(ii)	wave (with approx. same amplitude and) with correct phase difference		B1	[1]
6	(a)	(i)	movement/flow of charged particles	B1	[1]